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Bryan Donyanavard

Education

2013–2019 **Ph.D. in Computer Science**, *University of California, Irvine*.
Thesis: *Adaptive Resource Management for Mobile Multiprocessors through Computational Self-Awareness*

2008–2010 **M.Sc. in Computer Engineering**, *University of California, Santa Barbara*.

2004–2008 **B.Sc. in Computer Engineering**, *University of California, Santa Barbara*.

Research Interests

My research lies at the intersection of computer systems, embedded computing, and machine learning, with a focus on building runtime systems that adapt to changing conditions while meeting real-time, safety, and efficiency constraints. Three interconnected themes:

Adaptive Resource Management — Hierarchical, control-theoretic, and learning-based runtime management for heterogeneous multi-core systems-on-chip; coordinated control of task mapping, frequency, voltage, and memory hierarchies under performance and power targets.

Computational Self-Awareness — Systems that introspect on their own state, models, and goals at runtime to make adaptation decisions; supervisory and self-optimizing architectures for dependable embedded computing.

Autonomous Cyber-Physical Systems — On-device perception, SLAM, and neural-network inference for resource-constrained autonomous platforms; safety, predictability, and resilience in real-time autonomous systems.

Research & Work Experience

2021 – present **Assistant Professor**, Department of Computer Science, San Diego State University.
Founder and Director of the Computer Systems Intelligence Lab. Research focused on autonomous cyber-physical systems, on-device AI, and adaptive resource management. Currently supervising 7 students (1 PhD, 4 MS, 2 BS).

Summer 2024 **Visiting Researcher**, Center for Trustworthy Edge Computing Systems and Applications, KTH Royal Institute of Technology.
Collaborative research on edge computing for safety-critical autonomous systems.

2020 – 2021 **Experienced Researcher**, Cyber-Physical Systems, Ericsson Research.
Research on 5G-connected robotics and distributed SLAM, leading to two US patent applications and an IROS 2021 publication.

2019 – 2020 **Postdoctoral Researcher**, Center for Embedded and Cyber-physical Systems, University of California, Irvine.
Co-investigator on the Information Processing Factory (IPF) project; Reinforcement Learning–based runtime management for heterogeneous SoCs.

Postdoctoral Researcher, Chair of Integrated Systems, Technical University, Munich.

2015 – 2019 **Graduate Student Researcher**, Dutt Research Group, University of California, Irvine.
Doctoral research on hierarchical, self-aware resource management for mobile multi-core SoCs; lead developer of the open-source MARS framework.

Summer 2016 **Software Engineering Intern**, Chrome OS, Google.
Modeled NVM-based memory hierarchies for mobile SoCs in gem5; analyzed last-level cache controller policies.

Summer 2015 **Software Engineering Intern**, Technical Infrastructure, Google.
Memory-hierarchy analysis of live cloud workloads to evaluate multi-tiered, NUMA-aware memory management.

2011 – 2013 **Software Developer**, SPARC Systems Group, Oracle.
Production C firmware for SPARC server platforms (boot ROM, peripheral initialization) across three product lines.

2009 – 2010 **Engineer**, Special Technologies Laboratory, National Security Technologies.
RTOS-based embedded sensor devices and custom communication protocols for networked field deployment.

Honors & Awards

2024, 2026 **Assigned Time Award for Exceptional Service**, San Diego State University.
Recognition by the university for exceptional service contributions.

2013 – 2017 **ICS Dean's Fellowship**, University of California, Irvine.
Four years of full financial support for doctoral study.

Student Awards (under my supervision)

2026 **Top Undergraduate Student, CS Department**, San Diego State University.
Awarded to Jianna Gapuz (STEM Pathways and NSF REU advisee).

2026 **Runner Up Top Graduate Student, CS Department**, San Diego State University.
Departmental recognition for MS thesis advisee Changhoon Sung.

Teaching

2018 – present **Instructor**.

- Computer Science Capstone Project, San Diego State University
  - Fall 2025, Spring 2026
- Internet of Things, San Diego State University
  - Spring 2025, Spring 2023
- Intro to Software Systems, San Diego State University
  - Fall/Spring 2024, Fall/Spring 2023, Fall/Spring 2022, Fall/Spring 2021
- Digital Logic Design, UC Irvine
  - Fall 2018

2017 **Certificate in Teaching Excellence Program**, Division of Teaching Excellence and Innovation, University of California, Irvine.

2017 **Associate Training**, Center for the Integration of Research, Teaching and Learning, University of California, Irvine.

2014 – 2016 **Volunteer Tutor Lead**, Rocket Science Tutors, Santa Ana Unified School District.

Research Supervision & Mentoring

Current Research Advisees

PhD Alles Rebel, Changhoon Sung, Alyssa Serrano

MS Arshia Akhavan, Afnan Algharbi

Completed Theses & Mentored Projects

2026 **Summer Undergraduate Research Program**, *SDSU*, Tanner Kocher.

2026 **MS Thesis**, *SDSU*, Changhoon Sung.
A Study on Processing-near-memory-based Memory Compression Techniques for Improving Memory Efficiency in Autonomous Systems

2026 **NSF REU**, *SDSU*, Jianna Gapuz.

2026 **NSF REU**, *SDSU*, David Kaauwai.

2025 **MS Thesis**, *UniBw München*, Felix Gröte.
Reservoir Computing and its Implementation on an Edge Computer

2025 **MS Thesis**, *UniBw München*, Jacqueline Auner.
Cooperative Multi-Agent Mapping with Submaps

2025 **MS Thesis**, *UniBw München*, Fabian Bellgardt.
Migrating a Model Factory to the Cloud: AWS-Based Control Dashboard and Digital Twin

2025 **MESA Summer Program**, *SDSU*, Tuong Truong, Francisco Gomez.

2025 **STEM Pathways Summer Program**, *SDSU*, Jianna Gapuz.

2025 **BS Honors Thesis**, *SDSU*, Alyssa Serrano.

2024 **Summer Undergraduate Research Program**, *SDSU*, Alyssa Serrano, Adam Kaauwai.

2024 **STEM Pathways Summer Program**, *SDSU*, Jordan Kelley, Andres Cadena.

2023 **MS Thesis**, *UniBw München*, Marius Hillen.
Preprocessing images for gesture recognition in MediaPipe

2022 **STEM Pathways Summer Program**, *SDSU*, Fernando Quintana.

2022 **Summer Undergraduate Research Program**, *SDSU*, Lili Balazs, Christopher Fisher.

2021 **Summer Undergraduate Research Program**, *SDSU*, Nicholas Lozben.

2020 **MS Thesis**, *KTH*, Sandra Hernandez Herrero.
Cross-layer optimization for visual-inertial localization in resource constrained devices

2017 – 2018 **Mentor**, International Summer Undergraduate Research Fellowship, University of California, Irvine.

2010 **Graduate Student Mentor**, Apprentice Researchers Program, University of California, Santa Barbara.

Research Funding

2024 – 2028 **Expanding AI Innovation through Capacity Building and Partnerships (ExpandAI)**, *National Science Foundation, Division of Computing and Communication Foundations*.
PARTNER: Expanding AI Capacity in San Diego: A Strategic Collaboration between San Diego State University and TILOS AI Institute.
Role: Co-Principal Investigator. Total: $2,800,000 (Own share: $600,000).

2024 – 2026 **Computer and Information Science and Engineering Research Initiation Initiative (CRII)**, *National Science Foundation, Division of Computer and Network Systems*.
CRII: CNS: Supporting Resilient Perception in Autonomous Cyber-physical Systems.
Role: Principal Investigator. Amount: $193,000.

2024 **Seed Grant Program**, *San Diego State University*.
Enabling Flexible Autonomous Systems.
Role: Principal Investigator. Amount: $7,490.

Software & Open-Source Contributions

MARS, *Resource Management Policy Framework*, Lead developer / maintainer.
Open-source C++ runtime framework and offline simulator for adaptive resource management on mobile multi-core SoCs. Used by external research groups; basis of multiple peer-reviewed publications.
`https://github.com/duttresearchgroup/MARS`

Chauffeur, *Self-Driving Vehicle Benchmark Suite*, Contributor.
Benchmark suite for design and end-to-end analysis of self-driving vehicle pipelines on embedded systems. Published as a TECS journal article (2021).
`https://github.com/duttresearchgroup/Chauffeur`

gem5-spm, *Software-Programmable Memory Module for gem5*, Lead developer.
Extension to the gem5 architectural simulator adding support for software-programmable memories (SPMs), enabling reproducible research on alternative memory hierarchies.
`https://github.com/duttresearchgroup/gem5-spm`

Conference Papers

*Primary author † Primary supervisor

2025 Ermanno Bartoli, Rebecca Stower, Bryan Donyanavard, Hanna Werner, Jana Tumova, Iolanda Leite, *The Need for (Robot) Speed: Offloading Heavy Computations Improves Response Time and User Experience in Spoken Interactions*, IEEE International Conference on Robot and Human Interactive Communication (**RO-MAN** – *To appear*)

2025 Dongjoo Seo, Changhoon Sung, Ping-Xiang Chen, Bryan Donyanavard, Nikil Dutt, *SCHED: Safe CPU Scheduling Framework with Reinforcement Learning and Decision Trees for Autonomous Vehicles*, IEEE Vehicular Technology Conference: VTC2025-Spring Recent Results (**VTC**)

†2025 Matthew Bozoukov, Nguyen Anh Vu Doan, Bryan Donyanavard, *Generating and Predicting Output Perturbations in Image Segmenters*, Design, Automation & Test in Europe Conference & Exhibition (**DATE**)

†2025 Dongjoo Seo, Changhoon Sung, Junseok Park, Ping-Xiang Chen, Bryan Donyanavard, Nikil Dutt, *SPEED: Scalable and Predictable EnhancEments for Data Handling in Autonomous Systems*, International Symposium on Quality Electronic Design (**ISQED**)

2024 Danny Abraham, Biswadip Maity, Bryan Donyanavard, Nikil Dutt, *Back to the Future: Reversible Runtime Neural Network Pruning for Safe Autonomous Systems*, Design, Automation & Test in Europe Conference & Exhibition (**DATE**)

2022 Dongjoo Seo, Biswadip Maity, Ping-Xiang Chen, Dukyoung Yun, Bryan Donyanavard, Nikil Dutt, *ProSwap: Period-aware Proactive Swapping to Maximize Embedded Application Performance*, IEEE International Conference on Networking, Architecture and Storage (**NAS**)

2022 Sina Shahhosseini, Tianyi Hu, Dongjoo Seo, Anil Kanduri, Bryan Donyanavard, Amir Rahmani, Nikil Dutt, *Hybrid Learning for Orchestrating Deep Learning Inference in Multi-user Edge-cloud Networks*, International Symposium on Quality Electronic Design (**ISQED**)

†2021 Sandra Hernández, José Araujo, Patric Jensfelt, Ioannis Karagiannis, Ananya Muddukrishna, Bryan Donyanavard, *Cross-layer Configuration Optimization for Localization on Resource-constrained Devices*, International Conference on Intelligent Robots and Systems (**IROS**)

2020 Biswadip Maity, Bryan Donyanavard, Nikil Dutt, *Self-aware Memory Management for Emerging Energy-efficient Architectures*, 11th International Green and Sustainable Computing Workshops (**IGSC**)

2020 Florian Maurer, Bryan Donyanavard, Amir Rahmani, Nikil Dutt, Andreas Herkersdorf, *Emergent Control of MPSoC Operation by a Hierarchical Supervisor / Reinforcement Learning Approach*, Design, Automation & Test in Europe Conference & Exhibition (**DATE**)

*2019 Bryan Donyanavard, Armin Sadighi, Florian Maurer, Tiago Mück, Amir Rahmani, Andreas Herkersdorf, Nikil Dutt, *SOSA: Self-Optimizing Learning with Self-Adaptive Control for Hierarchical System-on-Chip Management*, 52nd IEEE/ACM International Symposium on Microarchitecture (**MICRO**)

2019 Biswadip Maity, Bryan Donyanavard, Nalini Venkatasubramanian, Nikil Dutt, *Workload Characterization for Memory Management in Emerging Embedded Platforms*, The 6th International Embedded Systems Symposium (**IESS**)

*2018 Bryan Donyanavard, Amir Mahdi Hosseini Mozannah, Tiago Mück, Nikil Dutt, *Exploring Hybrid Memory Caches in Chip Multiprocessors*, 13th International Symposium on Reconfigurable Communication-centric Systems-on-Chip (**ReCoSoC**)

*2018 Amir Rahmani, Bryan Donyanavard, Tiago Mück, Kasra Moazemmi, Axel Jantsch, Onur Mutlu, Nikil Dutt, *SPECTR: Formal Supervisory Control and Coordination for Many-core Systems Resource Management*, Proceedings of the Twenty-Third International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**)

*2018 Bryan Donyanavard, Amir Rahmani, Tiago Mück, Kasra Moazemmi, Nikil Dutt, *Gain Scheduled Control for Nonlinear Power Management in CMPs*, Design, Automation & Test in Europe Conference & Exhibition (**DATE**)

2017 Tiago Mück, Bryan Donyanavard, Nikil Dutt, *PoliCym: Rapid Prototyping of Resource Management Policies for HMPs*, Proceedings of the 28th International Symposium on Rapid System Prototyping: Shortening the Path from Specification to Prototype (**RSP**)

*2016 Bryan Donyanavard, Tiago Mück, Santanu Sarma, Nikil Dutt, *SPARTA: Runtime Task Allocation for Energy Efficient Heterogeneous Many-cores*, International Conference on Hardware/Software Codesign and System Synthesis (**CODES+ISSS**)

2016 Hossein Tajik, Bryan Donyanavard, Nikil Dutt, *On Detecting and Using Memory Phases in Multimedia Systems*, Proceedings of the 14th ACM/IEEE Symposium on Embedded Systems for Real-Time Multimedia (**ESTIMedia**)

2010 Yi-Chu Wang, Bryan Donyanavard, Tim Cheng, *Energy-Aware Real-Time Face Recognition System on Mobile CPU-GPU Platform*, European Conference on Computer Vision (**ECCV Workshops**)

Journal Articles

*Primary author † Primary supervisor

†2025 Alles Rebel, Nikil Dutt, Bryan Donyanavard, *OASIS: Optimized Adaptive System for Intelligent SLAM*, ACM Transactions on Embedded Computing Systems (**TECS**)

2025 Danny Abraham, Biswadip Maity, Bryan Donyanavard, Nikil Dutt, *Runtime Adaptivity for Efficient Neural Network Inference on Autonomous Systems*, ACM Transactions on Embedded Computing Systems (**TECS**)

2023 Florian Maurer, Moritz Thoma, Anmol Prakash Surhonne, Bryan Donyanavard, Andreas Herkersdorf, *Machine learning in run-time control of multicore processor systems*, **it - Information Technology**

2022 Sina Shahhosseini, DongJoo Seo, Anil Kanduri, Tianyi Hu, Sung-Soo Lim, Bryan Donyanavard, Amir Rahmani, Nikil Dutt, *Online Learning for Orchestration of Inference in Multi-User End-Edge-Cloud Networks*, ACM Transactions on Embedded Computing Systems (**TECS**)

2021 Biswadip Maity, Saehanseul Yi, Dongjoo Seo, Leming Cheng, Sung-Soo Lim, Jong-Chan Kim, Bryan Donyanavard, Nikil Dutt, *Chauffeur: Benchmark Suite for Design and End-to-End Analysis of Self-Driving Vehicles on Embedded Systems*, ACM Transactions on Embedded Computing Systems (**TECS**)

2021 Biswadip Maity, Bryan Donyanavard, et al., *SEAMS: Self-Optimizing Runtime Manager for Approximate Memory Hierarchies*, ACM Transactions on Embedded Computing Systems (**TECS**)

2020 Eberle Rambo, Bryan Donyanavard, Minjun Seo, Florian Maurer, et al., *The Self-Aware Information Processing Factory Paradigm for Mixed-Critical Multiprocessing*, **IEEE Transactions** on Emerging Topics in Computing

2020 Tianyi Zhang, Minjun Seo, Bryan Donyanavard, Nikil Dutt, Fadi Kurdahi, *Predicting Failures in Embedded Systems using Long Short-Term Inference*, IEEE Embedded Systems Letters (**ESL**)

2018 Tiago Mück, Bryan Donyanavard, Kasra Moazemmi, Amir Rahmani, Axel Jantsch, Nikil Dutt, *Design Methodology for Responsive and Robust MIMO Control of Heterogeneous Multicores*, IEEE Transactions on Multi-Scale Computing Systems (**TMSCS**)

2018 Majid Shoushtari, Bryan Donyanavard, Luis Angel D Bathen, Nikil Dutt, *ShaVe-ICE: Sharing Distributed Virtualized SPMs in Many-Core Embedded Systems*, ACM Transactions on Embedded Computing Systems (**TECS**)

2016 Aviral Shrivastava, Nikil Dutt, Jian Cai, Majid Shoushtari, Bryan Donyanavard, Hossein Tajik, *Automatic Management of Software Programmable Memories in Many-core Architectures*, **IET** Computers & Digital Techniques

2016 Hossein Tajik, Bryan Donyanavard, Janmartin Jahn, Joerg Henkel, Nikil Dutt, *SPMPool: Runtime SPM Management for Memory-Intensive Applications in Embedded Many-Cores*, ACM Transactions on Embedded Computing Systems (**TECS**)

Book Chapters

2021 Bryan Donyanavard, Amir M Rahmani, Axel Jantsch, Onur Mutlu, Nikil Dutt, **Intelligent Management of Mobile Systems Through Computational Self-Awareness**, *Handbook of Research on Methodologies and Applications of Supercomputing*

2020 Bryan Donyanavard et al., **Reflecting on Self-Aware Systems-on-Chip**, *A Journey of Embedded and Cyber-Physical Systems*

Patents

2024 **Updating a Map for Use in Localizing a Mobile Device**, *US Patent Application No. 2024/0310186 A1*, Inventors: Ioannis Karagiannis, José Araújo, Bryan Donyanavard

2024 **Predicting Performance of a Localization-Related Device**, *US Patent Application No. 2024/0015479 A1*, Inventors: José Araújo, Bryan Donyanavard, Ioannis Karagiannis, Ananya Muddukrishna

Service

Conference Organization	ESWEEK 2026 (Publication Co-Chair)
Workshop Organization	NetAISys 2025 (TPC Chair), SelPhyS 2024 (Organizer), TACPS 2024 (Local Organizer)
Committee	CODES+ISSS, DAC, GLSVLSI, ECRTS, NetAISys, ACSOS
Reviewer	ISCA, CASES, ESTIMedia, ACM SAC, ACM TODAES, CODES+ISSS, VLSI Design, SCOPES, DATE, IEEE TC, ACM TACO, GLSVLSI, IEEE TVLSI, IEEE ISVLSI, IEEE Design & Test, CCPE, ASP-DAC, ACM TECS